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Transforming healthcare through digital twin technology: Current evidence and emerging opportunities

Aanuoluwapo Clement David-Olawade [a,*], Racheal Boluwatife Abanishe [b], Muyiwa Ademola Ogunbona [c], Babajide David Makanjuola [d], Eniola Dorcas Oluwadairo [e], Omobolaji Rosemary Olasilola [f]

[a] Endoscopy Unit, Glenfield Hospital, University Hospitals of Leicester, NHS Trust, Leicester, United Kingdom
[b] Department of Medical Laboratory Science, Kwara State University, Ilorin, Nigeria
[c] Cell Biology and Genetics Unit, Department of Zoology, University of Ibadan, Ibadan, Nigeria
[d] Sheffield Business School, Department of Accounting, Banking and Finance, Sheffield Hallam University, Howard St, Sheffield City Centre, Sheffield S1 1WB, United Kingdom
[e] Faculty of Health and Life Sciences, Ulster University, London Campus, London, United Kingdom
[f] Department of Public Health, University of Hertfordshire, Hatfield AL10 9AB, United Kingdom

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ABSTRACT

Digital Twin (DT) technology, originating from engineering and manufacturing, has emerged as a transformative innovation in healthcare, offering unprecedented capabilities for real-time simulation, personalised medicine, and predictive analytics. A digital twin is a dynamic virtual representation of physical entities that continuously update through data integration, enabling advanced modelling and evidence-based decision-making. This narrative review aims to explore the conceptual foundations, clinical applications, benefits, challenges, and future directions of digital twin technology in healthcare, providing a comprehensive synthesis of current evidence and identifying knowledge gaps for future research. We conducted a narrative review of peer-reviewed literature published between 2010 and 2025. Multiple databases including PubMed, Scopus, Web of Science, IEEE Xplore, and Google Scholar were searched using terms related to digital twins, healthcare applications, precision medicine, and predictive analytics. Articles were selected based on relevance, quality, and contribution to understanding DT applications across diverse healthcare domains. Following the application of inclusion and exclusion criteria, a total of 74 articles were retained for synthesis. Digital twins demonstrate diverse applications spanning personalised medicine, surgical planning, chronic disease management, hospital operations, and rehabilitation. Evidence suggests improvements in treatment personalisation, predictive capabilities, risk reduction, and patient engagement. However, significant challenges persist, including technical barriers in data integration, ethical concerns regarding privacy and algorithmic bias, regulatory uncertainties, and clinical adoption barriers related to workflow integration and trust. Digital twin technology holds transformative potential for healthcare delivery, enabling predictive and preventive approaches. Realising this potential requires addressing technical, ethical, and regulatory challenges through interdisciplinary collaboration and responsible implementation frameworks.

1. Introduction

The healthcare sector stands at the cusp of a digital revolution, driven by the convergence of artificial intelligence (AI), the Internet of Things (IoT), big data analytics, cloud computing, and advanced sensor technologies. These technological advances are fundamentally reshaping how medical professionals diagnose diseases, deliver treatments, monitor patient outcomes, and manage healthcare systems.[1] Within this rapidly evolving landscape, Digital Twin (DT) technology has emerged as a particularly disruptive innovation with the potential to fundamentally transform patient care paradigms,[2] moving healthcare from reactive treatment models towards predictive, preventive, and

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* Corresponding author.
E-mail address: aanuclement23@gmail.com (A. Clement David-Olawade).

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truly personalised medicine.[3]

A digital twin, in its essence, is a sophisticated virtual representation of a physical system or entity that accurately mirrors its real-world counterpart through continuous, bidirectional data exchange. In the healthcare context, this translates to creating a living, breathing digital model of an organ, an individual patient, a medical device, or even an entire hospital system.[4,5] These models are not static snapshots but dynamic entities that evolve in real-time with clinical data from electronic health records (EHRs),[6] genetic information from genomic sequencing, physiological metrics from wearable sensors, imaging data from CT and MRI scans, and behavioural patterns from patient monitoring systems.[4] This continuous data integration provides a dynamic framework for simulation, prediction, and scenario testing that was previously impossible.[3]

The digital twin concept, first popularised by Michael Grieves in 2002 and subsequently applied extensively in aerospace engineering by NASA and in manufacturing by companies like General Electric and Siemens, is now finding its way into biomedicine and healthcare.[7] Technology has proven invaluable in industries where system failures carry catastrophic consequences from predicting aircraft engine failures to optimising wind turbine performance. Healthcare, where individual variability is immense and treatment decisions carry life-or-death implications, presents both a compelling use case and unique implementation challenges.[8] Recent advances in computational power, machine learning algorithms, and sensor miniaturisation have made healthcare-specific digital twins increasingly feasible and practical.[9]

Applications of DT in healthcare are diverse and expanding rapidly. In personalised medicine, digital twins enable clinicians to test treatment strategies virtually before administering them to patients, potentially reducing adverse effects and optimising therapeutic outcomes.[10] Surgeons are using patient-specific digital twins to rehearse complex operations, reducing operative time and complications. In chronic disease management, DT systems integrate continuous monitoring data to predict exacerbations and guide timely interventions.[11] At the systems level, hospital administrators employ digital twins to optimise patient flow, resource allocation, and capacity planning, and applications that proved particularly valuable during the COVID-19 pandemic when healthcare systems faced unprecedented strain.[12]

Despite this promising potential, significant challenges remain before digital twin technology can achieve widespread clinical adoption.[12] The statement of the problem centres on several interconnected issues: the technical complexity of integrating heterogeneous data sources, the computational intensity of real-time modelling, ethical concerns regarding patient data privacy and algorithmic bias, regulatory uncertainty surrounding clinical validation and approval pathways, and clinician hesitancy stemming from trust, liability, and workflow integration concerns. The rationale for this review emerges from the need to synthesise the rapidly growing but fragmented evidence base on healthcare DT applications, providing researchers, clinicians, and policymakers with a comprehensive understanding of current capabilities and limitations. The contribution of this narrative review lies in its broad synthesis of DT applications across multiple healthcare domains, from molecular to population levels, alongside a critical analysis of technical, ethical, and regulatory dimensions. Consistent with a narrative rather than a systematic methodology, this work does not claim exhaustive literature coverage or meta-analytic conclusions, but rather provides a structured, thematic synthesis of current evidence and identifies priority areas for future research. The aim of this narrative review is to systematically explore the current state and future potential of digital twin technology in healthcare. The specific objectives are: (1) to elucidate the conceptual framework and core components of healthcare digital twins; (2) to synthesise evidence on clinical applications across diverse healthcare settings; (3) to identify and analyse technical, ethical, and regulatory challenges; (4) to propose future research directions and implementation pathways; and (5) to assess the implications for healthcare delivery, patient outcomes, and system sustainability.

2. Methods

2.1. Review Design

This study employed a narrative review methodology to provide a comprehensive overview of digital twin technology in healthcare. Unlike systematic reviews that aim for exhaustive literature coverage with rigid protocols, narrative reviews allow for broader exploration of complex, multidisciplinary topics and synthesis of diverse evidence types including conceptual frameworks, empirical studies, and theoretical perspectives. This approach was deemed appropriate given the emerging nature of digital twin technology in healthcare, the heterogeneity of applications across clinical domains, and the need to integrate technical, clinical, ethical, and regulatory perspectives.

2.2. Search Strategy and Data Sources

A comprehensive literature search was conducted across multiple electronic databases to identify relevant publications. The databases searched included:

- PubMed/MEDLINE – for biomedical and clinical literature
- Scopus – for multidisciplinary peer-reviewed literature
- Web of Science – for scientific publications across disciplines
- IEEE Xplore – for engineering and technical literature
- Google Scholar – for grey literature and additional sources

The search covered publications from January 2010 to December 2025. This timeframe was selected because digital twin applications in healthcare began emerging significantly after 2010, following advances in computational power, sensor technologies, and artificial intelligence.

2.3. Search Terms and Boolean Operators

The following search strategy was employed using Boolean operators to combine key concepts:

Primary search string: ("digital twin*" OR "virtual twin*" OR "digital replica*" OR "virtual model*" OR "computational model*") AND ("healthcare" OR "medicine" OR "clinical" OR "patient care" OR "hospital*" OR "medical" OR "health system*")

Supplementary searches included specific application domains:

- Digital twin AND (personalised medicine OR precision medicine)
- Digital twin AND (surgery OR surgical planning OR operative)
- Digital twin AND (chronic disease OR disease management OR monitoring)
- Digital twin AND (hospital operations OR healthcare systems OR resource allocation)
- Digital twin AND (rehabilitation OR patient engagement)
- Digital twin AND (predictive analytics OR simulation OR modelling)

Search terms were adapted to each database's specific syntax and controlled vocabulary (e.g., MeSH terms in PubMed).

2.4. Inclusion and Exclusion Criteria

Inclusion criteria:

- Peer-reviewed journal articles, conference proceedings, and review papers
- Publications in English language
- Studies focusing on digital twin applications in human healthcare
- Articles discussing conceptual frameworks, technical implementations, clinical applications, or ethical/regulatory considerations
- Both theoretical/conceptual papers and empirical studies

Exclusion criteria:

- Articles published before 2010
- Non-English publications
- Studies focusing solely on veterinary medicine or agricultural applications
- Publications not accessible through institutional access or open access
- Abstracts without full-text availability
- Duplicate publications

2.5. Quality Assessment

Given the narrative nature of this review and the heterogeneity of included studies (ranging from conceptual papers to clinical trials), formal quality assessment using standardised checklists (such as CASP or Newcastle-Ottawa Scale) was not uniformly applicable. Instead, articles were evaluated based on:

- Peer-review status and journal reputation
- Methodological rigour appropriate to study design
- Clarity of reporting
- Relevance to review objectives
- Contribution to knowledge synthesis

Preference was given to recent publications (2020–2025), high-impact journals, and studies with robust methodologies, though earlier seminal works establishing conceptual foundations were also included. Where possible, priority was given to original primary research articles over secondary review citations, with reviews included where primary evidence was absent or where comprehensive synthesis of the field was required.

2.6. Data Extraction and Synthesis

Data extraction focused on capturing:

- Study characteristics (authors, year, country, study design)
- Digital twin application domain (personalised medicine, surgery, chronic disease, systems-level, etc.)
- Technical components and architecture
- Clinical outcomes and benefits reported
- Challenges and limitations identified
- Future directions proposed

Given the narrative review design, data synthesis was qualitative and thematic rather than quantitative meta-analysis. In total, 74 articles met the inclusion criteria and were retained for synthesis, comprising primary research studies, narrative and systematic reviews, and conceptual papers. Articles were organised by application domain and analysed to identify common themes, patterns, and knowledge gaps. Tables were constructed to synthesise key information across multiple studies, providing structured summaries of digital twin components, applications, and challenges.

3. Conceptual Framework of Digital Twin in Healthcare

3.1. Defining the Digital Twin

The term "digital twin" was first coined by Michael Grieves in 2002 during a presentation on product lifecycle management, though the underlying concept of digital representations has roots in NASA's Apollo programme, where engineers created physical duplicates of spacecraft systems for ground-based simulation and troubleshooting.[13] In the healthcare context, a digital twin represents a virtual, computational model of a physiological system, patient, or healthcare process that replicates its physical counterpart with sufficient fidelity to enable meaningful simulation, prediction, and optimisation.

The defining characteristic that distinguishes digital twins from traditional computational models or simulations is the bidirectional, real-time data flow between the physical entity and its virtual representation.[14] Whereas conventional models might be created once and run with hypothetical inputs, a digital twin continuously ingests data from its physical counterpart through sensors, medical devices, imaging systems, or health records, and updates its internal parameters accordingly.[15] This creates a dynamic model that evolves alongside the patient or system it represents, enabling more accurate predictions and timely interventions.

Healthcare digital twins can be conceptualised at multiple scales and levels of complexity. At the molecular level, twins might model protein folding, drug-receptor interactions, or genetic expression patterns. At the organ level, digital twins can simulate cardiac electrophysiology, pulmonary gas exchange, or hepatic metabolism.[16] At the patient level, comprehensive twins integrate multiple organ systems to predict individual treatment responses or disease trajectories. At the systems level, digital twins might model emergency department patient flow, surgical suite utilisation, or epidemic spread within communities.[16] This multi-scale nature presents both opportunities for holistic health management and challenges for data integration and computational tractability.

3.2. Core Components and Architecture

Digital twins in healthcare typically comprise five interconnected components that work in concert to enable their functionality.[10] First, data acquisition systems capture information from the physical entity through diverse sources: wearable biosensors measuring heart rate, blood glucose, or activity levels; medical imaging modalities like CT, MRI, or ultrasound providing anatomical and functional information; genomic sequencing platforms revealing genetic predispositions; electronic health records documenting clinical encounters, diagnoses, and treatments; and patient-reported outcomes capturing symptoms and quality of life. The richness and granularity of input data directly influence the accuracy and utility of the digital twin.[17,18]

Second, data integration and processing platforms aggregate these heterogeneous data streams, addressing challenges of interoperability, standardisation, and quality control.[2,19,20] This layer employs AI and machine learning algorithms for data cleaning, normalisation, feature extraction, and pattern recognition.[2,3,11] Natural language processing may extract insights from unstructured clinical notes, while computer vision algorithms interpret medical images. The integration layer transforms raw data into structured, actionable information that feeds the simulation models.[5]

Third, computational models and simulation engines form the core of the digital twin, representing the underlying physiology, pathology, or system dynamics. These may include mechanistic models based on first principles (such as finite element models of cardiac mechanics), empirical models derived from population data, or hybrid approaches combining physiological knowledge with data-driven machine learning. The sophistication of these models ranges from relatively simple statistical predictors to complex multi-physics simulations requiring high-performance computing resources.[3,20,21]

Fourth, visualisation and user interfaces present model outputs in clinically meaningful formats, enabling healthcare professionals and patients to interpret predictions, explore scenarios, and make informed decisions.[2] Advanced visualisation might include 3D anatomical renderings, interactive dashboards displaying risk predictions, or immersive virtual reality environments for surgical planning. The interface design critically influences clinical adoption, as overly complex presentations may hinder rather than facilitate decision-making.[22,23]

Fifth, feedback and control mechanisms close the loop by translating model insights into actions that affect the physical entity, whether

adjusting medication dosages, recommending lifestyle modifications, triggering clinical alerts, or optimising hospital resource allocation.[2,20] This bidirectional interaction distinguishes digital twins from passive monitoring systems, making them active participants in the care process.[5] Over time, as the twin observes the outcomes of interventions, machine learning algorithms can refine predictions through adaptive learning, improving accuracy with accumulated experience. It is important to note that DT should not be conceptualised solely as a machine learning system. Rather, it constitutes an integrated ensemble of interacting components, data acquisition, integration, mechanistic and/or data-driven simulation, visualisation, and feedback control, that together create a dynamic, bidirectional virtual replica. Machine learning is one enabling element of this architecture, not the defining feature; conflating ML models with DT frameworks risks obscuring the distinct value proposition that DT technology offers over standalone predictive algorithms. At a high level, the architecture of a healthcare digital twin can be understood through its five core components, as illustrated in Fig. 1.

This figure illustrates the five essential building blocks of healthcare digital twin systems: data acquisition, data integration and processing, simulation models, visualisation, and the bidirectional feedback-control loop. Together, these components enable dynamic, real-time virtual representations of patients, organs, or healthcare processes, forming the foundation for predictive analytics and personalised medicine.

Table 1 summarises the core components of healthcare digital twins, their functions, and representative technologies.

Table 1
Core Components of Healthcare Digital Twin Systems.

Component	Primary Function	Representative Technologies	Key Challenges
Data Acquisition [21,22]	Capture real-time physiological, clinical, and behavioural data	Wearables (Fitbit, Apple Watch), medical imaging (MRI, CT), genomic sequencers (Illumina), EHR systems (Epic, Cerner), IoMT devices	Sensor accuracy, data continuity, standardisation, patient adherence
Data Integration [2,23–25]	Aggregate, clean, and structure heterogeneous data streams	FHIR standards, HL7 protocols, cloud platforms (AWS, Azure), data lakes, ETL pipelines	Interoperability, data quality, latency, security
Simulation Models [26]	Replicate physiological or system behaviour for prediction	Finite element analysis, agent-based models, neural networks, physiological simulators (Living Heart Project)	Model validation, computational cost, biological complexity, parameter uncertainty
Visualisation [24,27,28]	Present insights in clinically actionable formats	3D rendering (3D Slicer), dashboards (Tableau), VR/AR interfaces, mobile applications	Cognitive load, interpretability, clinical workflow integration
Feedback Control [28]	Translate predictions into interventions	Clinical decision support systems, automated alerts, closed-loop drug delivery, hospital management systems	Trust, liability, regulatory approval, human oversight requirements

3.3. Theoretical Foundations

The theoretical underpinnings of healthcare digital twins draw from multiple disciplines, creating an inherently interdisciplinary framework. From systems biology, DT inherits the concept of biological systems as complex, interconnected networks where perturbations in one component cascade through multiple pathways.[24,29] From control theory and cybernetics, DT borrows feedback mechanisms and optimisation principles that enable self-adjusting systems. From computational science, DT leverages numerical methods, high-performance computing, and algorithmic advances that make complex simulations tractable.[3,26] From clinical medicine, DT incorporates evidence-based practice, diagnostic reasoning, and therapeutic principles that ground models in medical reality.

A crucial theoretical distinction exists between mechanistic (physics-based) and empirical (data-driven) approaches to digital twin construction. Mechanistic models, such as those used in cardiac electrophysiology,[3,24] rely on mathematical equations derived from first principles of biology and physics, for instance, modelling ion channel dynamics or blood flow using established physical laws. These models offer interpretability and can extrapolate beyond observed data, but require substantial physiological knowledge and computational resources. Empirical models, conversely, learn patterns directly from data using machine learning techniques, offering flexibility and scalability but potentially sacrificing interpretability and struggling with novel scenarios outside their training distributions.[22,26,30]

Increasingly, hybrid approaches combine mechanistic and empirical elements, leveraging physiological constraints to guide data-driven learning while using machine learning to estimate parameters that are difficult to measure directly. For example, a cardiac digital twin might use physics-based equations to model blood flow dynamics while employing neural networks to predict patient-specific valve properties

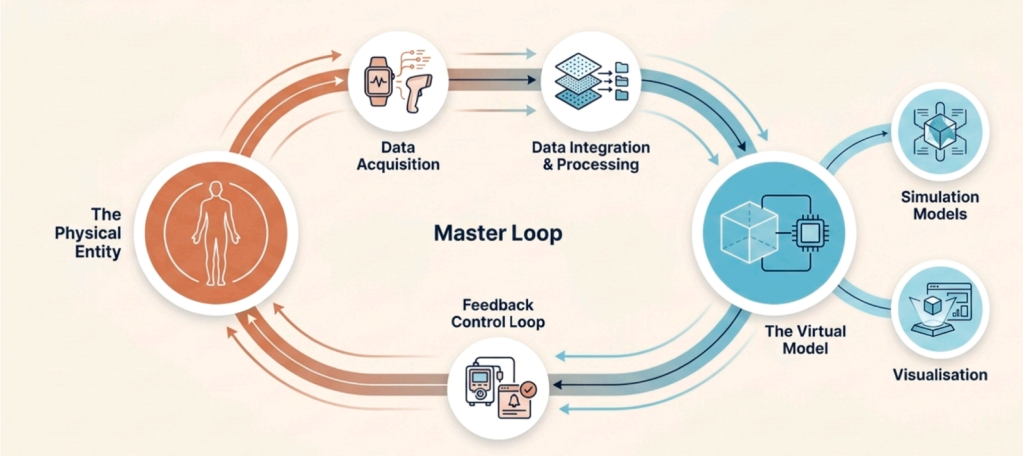


Fig. 1. Core architectural components of a healthcare digital twin.

from echocardiographic data.[24,28] This hybrid paradigm represents a promising middle ground that balances interpretability, accuracy, and computational efficiency. The trade-offs between these modelling approaches have direct implications for real-world deployment. Mechanistic models, while interpretable and theoretically grounded, impose significant computational latency that may preclude real-time clinical use without model reduction techniques such as surrogate modelling or order reduction. Empirical models are computationally faster but are vulnerable to performance degradation under data sparsity, distribution shift, and out-of-distribution inputs, conditions frequently encountered in clinical practice across diverse patient populations. Hybrid approaches partially mitigate these limitations but introduce complexity in model maintenance, validation, and regulatory approval.[31] The appropriate modelling strategy therefore depends on the specific clinical context: mechanistic models may be preferred where interpretability and safety-critical validation are paramount (e.g., cardiac electrophysiology), while data-driven approaches may be more suitable for high-volume, pattern-recognition tasks (e.g., patient flow prediction in hospital operations).[32]

4. Applications of Digital Twin in Healthcare

4.1. Personalised Medicine and Disease Modelling

Perhaps the most transformative application of digital twin technology lies in personalised medicine, where treatment decisions are tailored to individual patient characteristics rather than population averages. Traditional clinical practice relies heavily on randomised controlled trials that establish average treatment effects across populations, but individual patients often deviate substantially from these averages due to genetic variations, comorbidities, lifestyle factors, and environmental exposures.[20,33] Digital twins offer a pathway to precision medicine by creating patient-specific models that predict individual treatment responses before therapies are administered. To contextualise the evidence presented in this section and those that follow, applications are distinguished where possible by their stage of development: conceptual/theoretical, early-stage research or pilot implementation, ongoing investigation, or clinically implemented. This distinction is important given that many DT applications in healthcare remain at the pilot or investigational stage, and conflating research prototypes with validated clinical tools would misrepresent the maturity of the field.[34]

In cardiovascular medicine, digital twins of the human heart have demonstrated remarkable utility in predicting arrhythmias, optimising antiarrhythmic drug regimens, and planning catheter ablation procedures for atrial fibrillation.[3,30] Researchers at Johns Hopkins University developed personalised cardiac digital twins by integrating late-gadolinium enhancement MRI data, which reveals cardiac scar tissue, into computational models of cardiac electrophysiology.[20,30,35] These models successfully predicted which patients would benefit from implantable cardioverter-defibrillators and identified optimal ablation targets, potentially reducing procedure times and improving success rates while minimising tissue damage.[36] These cardiac digital twin studies remain predominantly at the research and early clinical validation stage, with most evidence derived from retrospective analyses and small prospective cohorts rather than large-scale randomised trials.

Oncology represents another frontier for digital twin applications, where tumour heterogeneity and treatment resistance pose formidable challenges. Researchers have developed digital twins that simulate tumour growth dynamics, vascular development, and responses to chemotherapy, radiotherapy, or immunotherapy.[37] By integrating imaging data, genomic profiles, and treatment histories, these models can predict which patients are likely to respond to specific therapies, potentially sparing non-responders from ineffective treatments and their associated toxicities. For instance, glioblastoma digital twins have been used to optimise radiotherapy planning by simulating radiation dose distributions and predicting tumour recurrence patterns, enabling clinicians to direct higher doses to high-risk regions while sparing healthy tissue.[38]

In diabetes management, digital twins integrate continuous glucose monitoring data with dietary logs, physical activity patterns, insulin dosing history, and individual metabolic characteristics to predict blood glucose trajectories hours or days in advance. This enables proactive adjustments to insulin regimens, dietary recommendations, and activity plans, potentially reducing both hypoglycaemic episodes and long-term complications.[39] Several research groups have developed "artificial pancreas" systems that use digital twin models to automatically adjust insulin delivery in real-time, achieving tighter glucose control than traditional pump therapy.[40]

Neurological applications are emerging as well, with digital twins being explored for epilepsy management, Parkinson's disease monitoring, and multiple sclerosis progression prediction. Virtual brain models can simulate seizure propagation patterns to identify optimal sites for surgical resection or deep brain stimulation, while wearable sensor data feeds into models predicting Parkinson's symptom fluctuations, enabling timely medication adjustments.[41,42] These neurological applications are predominantly at the early investigational stage, with virtual brain models for epilepsy demonstrating feasibility in small patient cohorts.[41] Digital twin technology is being rapidly adopted across multiple clinical and operational domains, as summarised in Fig. 2.

Two additional domains merit recognition in the context of personalised DT applications: radiology and drug discovery. In radiology, DT frameworks are being explored to integrate multi-modal imaging data, including CT, MRI, and PET, into computational models that can simulate disease progression or treatment response at the organ level, potentially enhancing diagnostic accuracy and reducing the need for repeat imaging.[26] These applications remain largely at the investigational stage. In drug discovery, DT models that simulate molecular-level interactions between candidate compounds and patient-specific biological targets offer the prospect of accelerating preclinical evaluation and reducing attrition in clinical trials; however, validated implementations in clinical drug development pipelines remain limited.[20,24] Across all personalised medicine applications, a critical consideration is the extent to which patient-specific data, including imaging, genomics, biomarkers, and longitudinal clinical records, are used to individualise the model, as this determines the degree of true personalisation achieved versus population-level approximation.

The figure summarises four major categories of digital twin implementation: personalised medicine, surgical planning, chronic disease management, and system-level hospital operations. Across these domains, digital twins support simulation, forecasting, and precision decision-making that improve outcomes and operational efficiency.

4.2. Surgical Planning and Training

Surgical specialities are increasingly adopting digital twin technology to enhance procedural planning, reduce operative risks, and train the next generation of surgeons.[43] Traditional surgical planning relies on static imaging studies, CT scans, MRIs, or X-rays, that provide anatomical information but limited functional insights. Digital twins extend this paradigm by creating interactive, patient-specific virtual models that surgeons can manipulate, test, and interrogate before making the first incision.[44]

In cardiac surgery, digital twins derived from 4D flow MRI and echocardiography enable surgeons to visualise complex congenital heart defects, simulate repair techniques, and predict post-operative haemodynamics.[45] For instance, in tetralogy of Fallot repair, surgeons can virtually test different patch configurations for ventricular septal defect closure, assessing how each option affects blood flow patterns and pressure gradients.[46,47] Similarly, for aortic valve replacement, digital twins help select appropriate prosthetic valve sizes and predict the risk of paravalvular leaks or conduction abnormalities requiring pacemaker implantation.

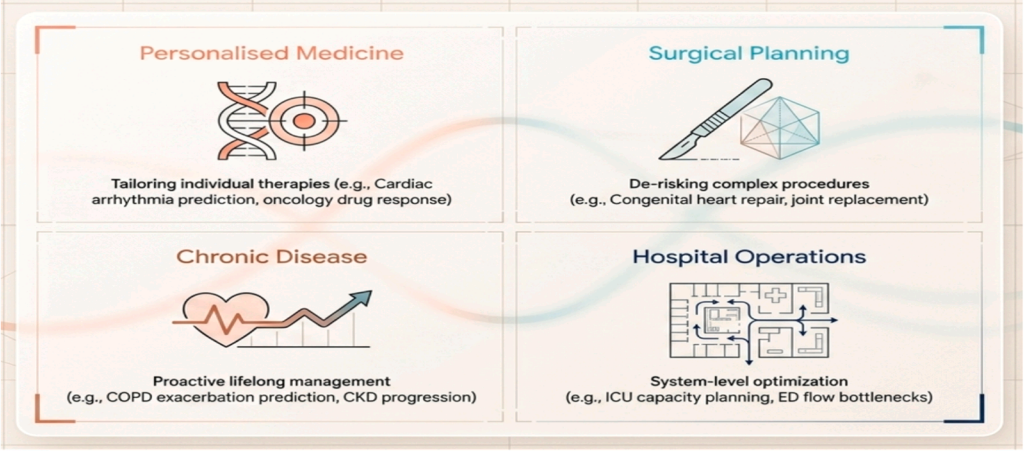


Fig. 2. Overview of key application domains of digital twin technology in healthcare.

Orthopaedic applications include virtual planning of joint replacements, spinal fusions, and fracture repairs.[48] A hip replacement digital twin might integrate CT imaging, gait analysis, and finite element modelling to optimise implant selection, positioning, and fixation, potentially improving longevity and functional outcomes while reducing dislocation risk. In complex spinal deformity corrections, surgeons can virtually test various rod configurations, screw placements, and osteotomy patterns, identifying approaches that achieve adequate correction while minimising stress concentrations that could lead to hardware failure.[46,48]

Neurosurgical applications are particularly compelling given the brain's functional complexity and the catastrophic consequences of surgical errors.[49] Digital twins can integrate structural imaging, functional MRI, diffusion tensor imaging (which maps white matter tracts), and electrocorticography data to create comprehensive models that map eloquent brain regions, areas critical for language, motor control, or vision, and predict how different surgical approaches might affect neurological function. For tumour resections, surgeons can virtually simulate extent-of-resection scenarios, balancing oncological goals against functional preservation.[50,51]

Beyond individual case planning, digital twins are transforming surgical education through immersive simulation environments.[52] Traditional surgical training follows an apprenticeship model where trainees gradually progress from observation to assisted performance under supervision. However, this model faces challenges from duty-hour restrictions, patient safety concerns, and the increasing complexity of minimally invasive techniques. Virtual reality surgical simulators powered by digital twin technology offer trainees unlimited opportunities for deliberate practice in risk-free environments.[44,52,53] These systems can simulate rare pathologies, unexpected complications, and high-stress scenarios that trainees might encounter infrequently in clinical practice, accelerating skill acquisition and improving preparedness.

4.3. Chronic Disease Management

Chronic diseases, including diabetes, chronic obstructive pulmonary disease (COPD), heart failure, chronic kidney disease, and hypertension, account for the majority of healthcare expenditure and morbidity in developed nations.[54] These conditions require ongoing monitoring, frequent treatment adjustments, and patient self-management over years or decades. Digital twins offer a paradigm shift from episodic, reactive care towards continuous, proactive management that anticipates exacerbations and optimises treatment before symptoms worsen.

In COPD management, digital twins integrate spirometry data, wearable sensor measurements (including activity levels, breathing patterns, and nocturnal oxygen saturation), environmental data (air quality, temperature, humidity), and patient-reported symptoms to predict exacerbations days before they become clinically apparent.[55] Early detection enables timely interventions, such as initiating corticosteroid bursts or adjusting bronchodilator regimens, that can prevent hospitalisations and emergency department visits. Some systems incorporate medication adherence data from smart inhalers, closing the loop between monitoring and intervention delivery.

Heart failure digital twins leverage data from implanted devices (such as cardiac resynchronisation therapy defibrillators that measure thoracic impedance as a proxy for fluid accumulation), wearable sensors, and laboratory values to predict decompensation events.[22,56] The CardioMEMS™ system, for instance, uses a wireless pulmonary artery pressure sensor combined with predictive algorithms to guide diuretic adjustments, reducing heart failure hospitalisations by approximately 30% in clinical trials.[57] It should be noted that the referenced 30% reduction in hospitalisations derives from the CHAMPION trial of the CardioMEM™ system, which evaluated a haemodynamic monitoring device rather than a fully realised digital twin per se; its inclusion here reflects the use of sensor-driven predictive algorithms that constitute a component of DT architecture.[57] Digital twin extensions of such systems might additionally incorporate echocardiographic parameters, biomarker trends, and patient activity patterns for more comprehensive risk stratification.

Chronic kidney disease (CKD) progression is notoriously variable, with some patients declining rapidly while others remain stable for decades. Digital twins that integrate longitudinal laboratory data (serum creatinine, estimated glomerular filtration rate, proteinuria), comorbidities, medications, and genetic factors can predict individual trajectories, identifying patients at high risk of progressing to end-stage renal disease requiring dialysis or transplantation. This enables risk-stratified care, directing intensive management towards high-risk patients while avoiding overtreatment of those likely to remain stable [20]It should be acknowledged that current evidence for DT-specific applications in CKD relies predominantly on general AI-based predictive modelling studies and secondary review sources [20]; dedicated primary research studies demonstrating validated DT implementations in CKD management remain limited and constitute an area requiring further investigation.[58]

Hypertension, while often perceived as straightforward to manage, exhibits substantial individual variability in treatment response. Digital twins that incorporate ambulatory blood pressure monitoring, medication adherence data, dietary sodium intake, physical activity levels, and genetic polymorphisms affecting drug metabolism can guide personalised antihypertensive regimens, potentially improving control rates and reducing cardiovascular events. The evidence base for DT applications in hypertension management is currently theoretical and extrapolated

from studies of ambulatory blood pressure monitoring and pharmacogenomics rather than from primary DT studies; prospective clinical trials are required before definitive claims about outcomes can be made.

A common theme across chronic disease applications is the integration of patient-generated health data from wearables and mobile applications with traditional clinical data. This continuous data stream enables digital twins to detect subtle changes that might be missed during infrequent clinic visits, facilitating timely interventions while reducing the burden of frequent in-person appointments, a particularly valuable capability highlighted during the COVID-19 pandemic when in-person care was disrupted. While DT technology was not itself directly deployed in the management of individual COVID-19 patients, the pandemic illustrated the value of continuous remote monitoring principles that underpin DT frameworks.

4.4. Hospital Operations and Healthcare Systems

Beyond individual patient care, digital twins are being deployed at the systems level to optimise hospital operations, resource allocation, and population health management. Healthcare delivery systems are extraordinarily complex socio-technical systems involving multiple stakeholders, competing priorities, stochastic arrival patterns, and resource constraints. System-level inefficiencies manifest as emergency department overcrowding, surgical cancellations, intensive care unit capacity crises, and staff burnout problems that digital twins can help address through simulation and optimization.[59]

Emergency department (ED) digital twins model patient arrival patterns, triage processes, diagnostic workflows, and disposition decisions to identify bottlenecks and test interventions virtually before implementation.[60] For instance, an ED twin might reveal that radiology capacity, rather than physician availability, limits throughput during evening hours, suggesting that adding an evening CT technician would reduce waiting times more effectively than adding another physician [61] Such insights, derived from simulation experiments that would be infeasible or unethical to conduct in live clinical settings, enable evidence-based operational improvements.

Surgical suite digital twins optimise scheduling, staffing, and resource allocation across operating rooms. These models incorporate case durations (with appropriate variability), turnover times, surgeon preferences, equipment availability, and staffing constraints to generate schedules that maximise utilisation while minimising overruns and evening finish times, a key determinant of staff satisfaction.[62] By simulating different scheduling policies, hospitals can identify approaches that balance efficiency, surgeon satisfaction, and patient access.

Intensive care unit (ICU) digital twins proved particularly valuable during the COVID-19 pandemic, when hospitals faced unprecedented surges in critically ill patients requiring mechanical ventilation.[63] Simulation and predictive modelling tools, drawing on principles central to DT frameworks, modelled admission flows, length-of-stay distributions, and resource requirements (ventilators, ICU beds, nursing staff, sedatives, neuromuscular blockers) to predict capacity shortfalls days or weeks in advance.[64] It should be noted, however, that the reference cited [63] describes outcomes of continuous positive airway pressure (CPAP) therapy in COVID-19 patients rather than the deployment of digital twin systems; where cited in the context of DT applications, readers should consult primary DT-specific sources such as.[65,66] This enabled proactive measures such as postponing elective surgeries, expanding ICU capacity into alternative spaces, negotiating mutual aid agreements with neighbouring hospitals, and securing additional equipment and staff.[67] Several countries developed national-level ICU digital twins to guide resource allocation across regions.

Supply chain digital twins help hospitals manage inventory of medications, medical devices, and personal protective equipment.[68] These models integrate consumption patterns, lead times from suppliers, stock levels, and demand forecasts to optimise ordering strategies, reducing both stockouts (which can compromise patient care) and excess inventory (which ties up capital). During supply chain disruptions, such as those experienced during the pandemic, digital twins can evaluate alternative sourcing strategies and substitution options.

Population health digital twins operate at community or regional scales, modelling disease prevalence, social determinants of health, healthcare utilisation patterns, and intervention impacts.[69,70] These models inform public health planning, resource allocation across geographic areas, and policy decisions regarding preventive programmes.[12] For example, a diabetes digital twin for a metropolitan area might integrate data on prevalence, demographic factors, food environments, built environments affecting physical activity, and healthcare access to identify neighbourhoods where targeted screening or intervention programmes would yield the greatest impact.

Table 2 summarises key applications of digital twin technology across different healthcare domains.

4.5. Rehabilitation and Patient Engagement

Rehabilitation following injury, surgery, or neurological events represents another promising application area for digital twin technology.[69,75] It is important to note that DT applications in rehabilitation are largely at the investigational and pilot stage; while the principles of personalised monitoring and adaptive exercise prescription are well established, fully realised DT systems, incorporating real-time bidirectional data flow, physiological simulation, and automated feedback, have not yet achieved widespread clinical validation or deployment in this domain. Traditional rehabilitation relies on periodic sessions with physical or occupational therapists who assess progress and adjust exercise programmes. However, the intermittent nature of these sessions may miss opportunities for optimisation, and patient adherence to home exercise programmes is often suboptimal due to lack of feedback and motivation.[75]

Digital twins in rehabilitation integrate biomechanical data from motion capture systems or wearable sensors with patient-reported outcomes, clinical assessments, and imaging studies to create personalised recovery models.[74] For post-operative anterior cruciate ligament (ACL) reconstruction rehabilitation, a digital twin might track knee range of motion, gait patterns, quadriceps strength, and proprioception to predict readiness for return-to-sport while assessing re-injury risk. The model can generate adaptive exercise programmes that progress based on actual performance rather than fixed timelines, potentially reducing re-injury rates while accelerating safe return to activity.

Stroke rehabilitation digital twins incorporate motor function assessments, neuroimaging data showing lesion location and brain reorganisation, and daily activity monitoring to predict recovery trajectories and optimise therapy intensity and focus. These models might identify patients likely to benefit from intensive constraint-induced movement therapy versus those for whom compensatory strategies would be more effective.[5,69] Virtual reality environments powered by digital twin technology can provide engaging, game-like rehabilitation exercises with real-time performance feedback, improving adherence and potentially enhancing neuroplasticity through enriched sensory-motor experiences.

An often-overlooked benefit of digital twin technology is enhanced patient engagement and health literacy.[16,24] By visualising their own digital twin, whether a 3D model of their heart, a graph showing predicted disease trajectory, or an interactive simulation of treatment options, patients gain deeper understanding of their condition and treatment rationale. This transparency can improve informed consent processes, increase treatment adherence, and empower patients to become active participants in their care rather than passive recipients. Several studies have demonstrated that patients who interact with visual representations of their health data show improved self-management behaviours and clinical outcomes.[24,69]

Table 2
Digital Twin Applications Across Healthcare Domains.

Application Domain	Specific Use Cases	Data Sources	Key Benefits	Implementation Examples
Personalised Medicine [3,28,34]	Cardiac arrhythmia prediction, cancer treatment selection, diabetes management, drug response prediction	MRI, genomics, continuous glucose monitors, wearables, EHRs	Individualised treatment, reduced adverse effects, improved outcomes	Living Heart Project (Dassault Systèmes), Hopkins cardiac DT, artificial pancreas systems
Surgical Planning [3,71]	Congenital heart repair, joint replacement, tumour resection, spinal surgery	CT, MRI, 4D flow imaging, functional MRI, biomechanical testing	Reduced operative time, fewer complications, optimised outcomes	Virtual Physiological Human project, FEniCS-HPC cardiac models
Chronic Disease [20,72]	COPD exacerbation prediction, heart failure monitoring, CKD progression, hypertension optimisation	Spirometry, implanted sensors (CardioMEMS), smart inhalers, wearables, labs	Earlier intervention, reduced hospitalisations, improved quality of life	CardioMEMS HF System, Propeller Health (COPD), Livongo (diabetes)
Hospital Operations [60,64,68]	ED flow optimisation, surgical scheduling, ICU capacity planning, supply chain management	Admission/discharge data, case durations, resource utilisation, patient flow patterns	Increased efficiency, reduced waiting times, better resource utilisation, cost savings	GE Healthcare Command Center, Philips Hospital to Home, COVID-19 ICU twins
Surgical Training [53,59]	VR simulation, rare pathology practice, complication management, technical skill development	Patient-specific imaging, physiological parameters, haptic feedback	Accelerated learning, improved competency, reduced training risks	Simbionix simulators, Osso VR, Touch Surgery platform
Rehabilitation [73,74]	Personalised exercise programmes, adherence monitoring, recovery prediction	Motion capture, wearables, patient-reported outcomes	Improved outcomes, patient engagement, optimised recovery trajectories	Hinge Health, Sword Health, Kaia Health

5. Benefits of Digital Twin Technology in Healthcare

5.1. Enhanced Personalisation

The most frequently cited benefit of digital twin technology is its ability to enable truly personalised medicine that moves beyond population-level evidence to individual predictions [24,32] The traditional paradigm of evidence-based medicine, while enormously valuable, has limitations when applied to individual patients who may differ substantially from trial populations in age, comorbidities, genetics, or social circumstances. Digital twins address this limitation by creating patient-specific models that account for individual variability, potentially improving treatment efficacy while reducing adverse effects and healthcare costs associated with trial-and-error approaches.[3,32]

5.2. Predictive and Preventive Capabilities

Digital twins shift healthcare from reactive to proactive by anticipating problems before they manifest clinically. Whether predicting heart failure exacerbations, surgical complications, or ICU capacity shortfalls, this predictive capability enables earlier interventions that may be less invasive, less expensive, and more effective than treating established problems.[24,25] The value proposition is particularly compelling for chronic diseases where early intervention can prevent irreversible organ damage, and for healthcare systems where proactive capacity planning can prevent crises.

5.3. Risk Mitigation

By enabling virtual experimentation, whether testing surgical approaches, medication regimens, or system-level policies, digital twins reduce risks associated with innovation and learning.[3,71] Surgeons can rehearse procedures on patient-specific models before operating, clinicians can virtually assess treatment responses before administering therapies, and administrators can evaluate operational changes before implementation. This de-risking is particularly valuable in high-stakes scenarios where errors carry severe consequences.

5.4. Economic Efficiency

Although digital twin implementation requires upfront investment in infrastructure, sensors, and computational resources, potential downstream savings may be substantial. Reduced hospitalisations through early intervention, shorter surgical times through better planning, optimised resource utilisation through simulation, and prevention of adverse drug reactions through virtual testing all contribute to the economic case for digital twins.[31,60,76] Several health economic analyses suggest that digital twins could deliver positive returns on investment, though real-world validation across diverse settings remains limited.

5.5. Improved Patient Engagement and Shared Decision-Making

Digital twins facilitate shared decision-making by making treatment options and their expected outcomes more tangible and comprehensible to patients.[24,32] Rather than discussing abstract statistics, clinicians can show patients their personalised digital twin and demonstrate how different treatments might affect their trajectory. This visualisation can improve health literacy, strengthen therapeutic alliances, and align treatment decisions with patient preferences and values.

6. Challenges and Limitations

6.1. Technical Challenges

Despite significant progress, formidable technical hurdles remain before digital twins can achieve widespread clinical deployment. Data integration represents perhaps the most immediate challenge, as healthcare data reside in fragmented systems with incompatible formats, terminologies, and access protocols.[77] Integrating data from electronic health records, medical imaging archives, laboratory information systems, genomic databases, wearable devices, and patient-reported outcome platforms requires robust interoperability standards, such as Fast Healthcare Interoperability Resources (FHIR), that remain incompletely implemented across healthcare systems.

Data quality poses another significant barrier. Missing data, measurement errors, temporal inconsistencies, and documentation variability can undermine model accuracy.[24,78] Wearable sensors may suffer from motion artefacts, skin contact issues, or calibration drift. Electronic health records contain unstructured text requiring natural language processing, coded diagnoses that may not reflect clinical reality, and systematically missing information (such as social determinants of health or over-the-counter medication use). Addressing these quality issues requires sophisticated data cleaning, imputation, and validation methods that remain active areas of research.[78]

Computational demands for real-time, high-fidelity simulation can be prohibitive. Detailed cardiac electrophysiology models, for instance, may require hours of computation on high-performance computing clusters for a single heartbeat simulation, making real-time clinical applications challenging without model reduction techniques or specialised hardware.[3,78] Balancing model complexity, computational

tractability, and prediction accuracy requires careful trade-offs that may vary across applications and clinical contexts.

Model validation and uncertainty quantification represent critical challenges that directly affect clinical trust and adoption. How do we validate a digital twin when the "ground truth" is often unknown or unmeasurable? Clinical trials typically require years of follow-up and thousands of patients to establish treatment efficacy, but digital twins make predictions for individual patients over short time horizons. Developing validation frameworks appropriate for personalised predictions, quantifying prediction uncertainty, and determining acceptable accuracy thresholds for clinical decision support are ongoing challenges requiring collaboration between clinicians, data scientists, and regulators.[24,78]

Biological complexity and individual variability limit model generalisability. Human physiology involves countless interacting processes spanning multiple temporal and spatial scales, from molecular signalling occurring in milliseconds to disease progression unfolding over decades.[79] Current models necessarily simplify this complexity, raising questions about their applicability across diverse patient populations, disease phenotypes, and clinical contexts. Models trained on relatively homogeneous populations may perform poorly when applied to different demographic groups, potentially exacerbating rather than ameliorating health disparities.

6.2. Ethical and Legal Concerns

Digital twin technology raises profound ethical questions that healthcare systems, professional societies, and regulators are only beginning to grapple with. Before examining these challenges, it is important to clarify that several of the concerns discussed in this section, including privacy, algorithmic bias, governance, and regulatory compliance, are not unique to digital twins. Many of these issues apply broadly to any AI-based predictive model used in clinical decision support. Where challenges are shared with the wider AI/ML landscape, this will be made explicit; and where challenges are specific or particularly acute for DT technology (such as the complexity of dynamic, continuously learning twin models or the need to manage bidirectional physical-virtual data flows in real time), these distinctions will be highlighted.[80] Data privacy concerns are paramount, as digital twins require continuous access to intimate health information spanning genomic data, physiological measurements, lifestyle behaviours, and clinical histories.[81] The data requirements for digital twins far exceed those of traditional healthcare IT systems, creating larger attack surfaces for breaches and more comprehensive surveillance capabilities.[82,83] Robust cybersecurity measures, including encryption, access controls, audit trails, and federated learning approaches that train models without centralising raw data, are essential but technically challenging.

Ownership questions arise: who owns the digital twin, the patient, the healthcare provider, the technology vendor, the insurer? Does a patient have the right to access, modify, or delete their digital twin? Can patients demand that predictions be incorporated into clinical decisions, or decline to have a digital twin created? These questions lack clear legal frameworks and may have different answers across jurisdictions with varying patient rights and data protection regulations.[24,84,85] These ownership questions are not unique to DT systems, they arise across healthcare AI more broadly, but they are particularly acute for DTs given the continuous, longitudinal, and multi-source nature of twin data, which creates a richer and more sensitive profile than most conventional AI systems.

Algorithmic bias represents a serious concern for healthcare equity. If digital twins are trained primarily on data from well-resourced healthcare systems serving relatively affluent populations, their predictions may be less accurate or entirely inappropriate for underserved communities. Historical patterns of healthcare disparities encoded in training data may be perpetuated or amplified by algorithmic predictions. For instance, an algorithm predicting healthcare resource needs trained on data from a system with unequal access might learn to allocate fewer resources to disadvantaged populations, reinforcing existing inequities. Ensuring fairness requires diverse training data, bias auditing, equity-focused validation, and ongoing monitoring for disparate impacts across demographic groups.[83,86] These algorithmic bias concerns are shared with all predictive healthcare AI systems; what distinguishes DT implementations is that bias may propagate and compound over time through the twin's adaptive learning mechanisms, potentially amplifying initial disparities as the model continues to update from biased data streams.

Informed consent processes need rethinking for digital twin applications. Traditional consent frameworks assume discrete interventions with defined risks and benefits, but digital twins are dynamic, continuously learning systems whose future behaviours may be difficult to predict at the time of initial consent.[87] How can patients meaningfully consent to algorithms that may change over time as they learn from population data? Should patients be re-consented when models are updated? These questions require new consent paradigms balancing patient autonomy, practicl feasibility, and ethical principles.

Liability concerns may slow adoption. If a digital twin predicts low risk of complications and a patient subsequently experiences adverse outcomes, who bears responsibility, the clinician who relied on the prediction, the institution that deployed the technology, or the vendor that developed the model? Conversely, if a clinician disregards a digital twin prediction and the patient suffers harm, does the twin's existence alter the standard of care? These liability questions remain largely unresolved and may require legislative clarification or novel insurance mechanisms to protect patients, clinicians, and innovators appropriately.[85]

6.3. Regulatory and Clinical Adoption Barriers

Regulatory pathways for digital twin approval remain unclear in most jurisdictions. Traditional medical device regulations were developed for static products with defined indications and fixed performance characteristics.[88] Digital twins challenge these frameworks because they continuously update through machine learning, raising questions about when revalidation is required and how post-market surveillance should be conducted. The US Food and Drug Administration's efforts to develop frameworks for software as a medical device (SaMD) and continuously learning algorithms represent important steps, but many uncertainties remain regarding evidentiary standards, approval timelines, and regulatory oversight intensity.[89,90]

The European Union's Medical Device Regulation (MDR) and In Vitro Diagnostic Regulation (IVDR), implemented in recent years, impose stringent requirements for clinical evidence that may prove challenging for digital twin technologies, particularly those making individualised predictions that cannot be validated through traditional clinical trial designs.[91,92] Balancing patient safety with innovation requires regulatory flexibility and ongoing dialogue between regulators, industry, clinicians, and patient advocates.

Clinical workflow integration presents practical barriers to adoption. Healthcare professionals already face overwhelming documentation burdens and alert fatigue from existing decision support systems.[93] Adding digital twin predictions to clinical workflows risks being ignored or actively resisted unless interfaces are intuitive, predictions are actionable, and the technology demonstrably improves efficiency rather than adding additional tasks. Successful implementation requires human-centred design, iterative refinement based on user feedback, and organisational commitment to workflow redesign rather than simply inserting new technology into existing processes.

Trust represents perhaps the most significant barrier to clinical adoption. Clinicians may be reluctant to rely on algorithmic predictions they don't fully understand, particularly when those predictions contradict clinical intuition or traditional guidelines.[86,94] Building trust requires transparency about model inputs, logic, and uncertainty;

validation in local patient populations demonstrating that predictions are reliable; and clinical champions who advocate for the technology based on positive experiences. The "black box" nature of some machine learning approaches may hinder trust-building, strengthening the case for interpretable models and explainable AI techniques that provide rationales for predictions.

Interestingly, among the well-validated healthcare AI applications to date have been those where algorithms handle well-defined, high-volume tasks (such as diabetic retinopathy screening or ECG interpretation) rather than comprehensive patient-level predictions. This suggests that incremental deployment of digital twin components addressing specific clinical needs may be more feasible than attempting to create comprehensive digital twins immediately.

Table 3 summarises the key challenges facing digital twin implementation across technical, ethical, and regulatory domains.[12,95]

Table 3
Key Challenges in Digital Twin Implementation.

Challenge Category	Specific Issues	Contributing Factors	Potential Solutions
Technical [12,19]	Data integration, computational demands, model validation, biological complexity	Fragmented IT systems, incompatible standards, high dimensionality, limited compute resources	FHIR adoption, cloud computing, federated learning, model reduction techniques, standardised validation frameworks
Data Quality [12,96]	Missing data, measurement errors, documentation variability, sensor reliability	Manual data entry, interoperability gaps, device limitations, workflow constraints	Automated data capture, IoMT devices, NLP for unstructured data, data quality metrics, imputation methods
Privacy & Security [12]	Data breaches, unauthorised access, surveillance capabilities, re-identification risk	Large data volumes, multiple access points, insider threats, sophisticated attacks	End-to-end encryption, federated learning, differential privacy, blockchain, zero-trust architecture, regular audits
Ethical [12,19]	Algorithmic bias, ownership disputes, consent challenges, equity concerns	Non-representative training data, legal ambiguity, inadequate frameworks, historical disparities	Diverse data collection, bias auditing, patient data rights, equity-focused design, inclusive governance
Regulatory [12]	Unclear approval pathways, continuously learning systems, post-market surveillance	Novel technology, adaptive algorithms, jurisdictional variations, legacy frameworks	Regulatory sandboxes, adaptive pathways, SaMD frameworks, international harmonisation, evidence standards
Clinical Adoption [12]	Workflow integration, alert fatigue, trust barriers, interpretability	Usability issues, cognitive load, opaque models, organisational resistance	Human-centred design, explainable AI, clinical validation, champions, training programmes, workflow co-design
Economic [12]	Implementation costs, reimbursement uncertainty, ROI validation	Infrastructure investments, maintenance costs, unclear value propositions	Health economic analyses, value-based payment models, shared savings agreements, cost-effectiveness trials

7. Future Directions

7.1. Multi-Scale Integration

The future trajectory of healthcare digital twins points towards increasingly comprehensive, multi-scale models that integrate molecular, cellular, organ, patient, and population levels within unified frameworks.[24] Current digital twins typically focus on single organs or specific physiological systems, but human health emerges from complex interactions across multiple systems and scales.[97] Future twins might integrate genomic and proteomic data predicting disease susceptibility, cellular models capturing tissue responses to injury or infection, organ-level models simulating physiological function, whole-body models accounting for inter-organ crosstalk, and population models contextualising individual risk within community epidemiology.[98]

The Virtual Physiological Human initiative, a European Commission-funded effort, exemplifies this multi-scale ambition by developing computational models spanning biological scales from genes to populations. Similarly, the Physiome Project aims to create comprehensive, integrative models of human physiology based on biophysical principles.[99] As these efforts mature and computational power continues to increase following Moore's Law trends, truly holistic digital twins encompassing biological complexity become increasingly feasible.[100]

7.2. Interoperability and Standardisation

Realising the full potential of digital twins requires robust interoperability standards enabling seamless data exchange across platforms, institutions, and national boundaries.[101] Current efforts around Fast Healthcare Interoperability Resources (FHIR), Observational Medical Outcomes Partnership (OMOP) common data models, and DICOM standards for medical imaging represent important foundations, but healthcare interoperability remains frustratingly incomplete. Future progress requires not only technical standards but also governance frameworks, financial incentives, and cultural shifts prioritising data sharing and collaboration over data hoarding and competitive advantage.

International collaboration through organisations like the International Medical Informatics Association (IMIA), Health Level Seven International (HL7), and the International Organization for Standardization (ISO) can accelerate standardisation while ensuring that standards reflect diverse healthcare system needs and cultural contexts. Public-private partnerships leveraging the agility of technology companies with the patient-centredness of healthcare providers may prove particularly effective in developing and disseminating standards [102]

7.3. Explainable and Trustworthy AI

As digital twins increasingly incorporate sophisticated machine learning models, the need for explainability and transparency becomes paramount. Explainable AI (XAI) techniques aim to make algorithmic predictions interpretable and actionable for clinicians without requiring deep technical expertise in machine learning. Approaches such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), attention mechanisms in deep learning, and counterfactual explanations offer promising pathways towards transparent digital twins that not only make predictions but explain their reasoning.[103]

Beyond technical explainability, building trustworthy digital twins requires addressing fairness, robustness, and reliability through rigorous testing, diverse validation datasets, and transparent reporting of performance across demographic subgroups and clinical contexts.[104] Organisations like the Coalition for Health AI (CHAI) are developing assurance frameworks and best practices for trustworthy healthcare AI, work that will be essential for digital twin adoption.

7.4. Regulatory Innovation

Regulatory frameworks must evolve to accommodate the unique characteristics of digital twin technology while maintaining rigorous safety and efficacy standards. Adaptive licensing pathways that allow initial market entry based on promising early evidence followed by post-market monitoring and iterative approval extensions may be appropriate for digital twins that continuously improve through learning. Regulatory sandboxes or innovation offices, already implemented in jurisdictions like the UK and Singapore, provide safe spaces for testing novel technologies under regulatory supervision before full market deployment.[105]

International regulatory harmonisation through initiatives like the International Medical Device Regulators Forum (IMDRF) can reduce duplication, accelerate innovation, and facilitate global deployment of validated digital twin technologies. However, harmonisation must respect legitimate differences in healthcare priorities, risk tolerances, and ethical values across societies.

7.5. Population-Level and Public Health Applications

While much digital twin development focuses on individual patient care, population-level applications offer compelling opportunities for public health impact. Digital twins of communities, cities, or nations could inform policy decisions about healthcare resource allocation, disease prevention strategies, environmental health interventions, and health equity initiatives.[25] During infectious disease outbreaks, population twins could guide testing strategies, contact tracing efforts, and non-pharmaceutical interventions by simulating epidemic dynamics under various scenarios.

Climate change adaptation represents another frontier for population digital twins. As extreme weather events, heat waves, and environmental shifts increasingly affect human health, models integrating climate projections, demographic trends, health system capacities, and population vulnerabilities could inform adaptation planning and resilience building. Such applications require interdisciplinary collaboration bridging public health, climate science, urban planning, and policy analysis [106]

7.6. Integration with Emerging Technologies

Digital twin technology will likely converge with other emerging healthcare innovations, creating synergies that amplify impact. Integration with blockchain technology could provide secure, auditable records of data provenance and model decisions while enabling patients to maintain sovereignty over their health data.[74] Quantum computing, as it matures beyond current experimental stages, may enable molecular-scale simulations currently computationally intractable, advancing precision medicine through accurate drug-protein interaction predictions. Neuromorphic computing architectures inspired by biological neural networks could provide energy-efficient platforms for edge computing in wearable digital twins.

The combination of digital twins with advanced robotics and autonomous systems could enable closed-loop medical devices that continuously optimise therapy delivery, such as artificial pancreas systems that adjust insulin in real-time or smart prosthetics that adapt to user intent. Integration with virtual and augmented reality could enhance surgical planning, patient education, and rehabilitation through immersive, interactive experiences grounded in patient-specific digital models.[107]

8. Implications for Healthcare Delivery and Health Systems

The widespread adoption of digital twin technology would fundamentally reshape healthcare delivery models, clinical roles, and health system economics. Traditional episodic care models organised around clinic visits and hospital admissions would likely shift towards continuous monitoring and proactive intervention, enabled by digital twins that provide always-on surveillance and prediction.[108] This transition raises questions about reimbursement models, as fee-for-service payment that rewards volume of encounters may not appropriately incentivise continuous digital health management. Value-based payment models that reward health outcomes or quality metrics may better align with digital twin capabilities, though developing appropriate metrics and attribution models remains challenging.[109]

Clinical roles may evolve as digital twins assume certain analytical and predictive tasks. Rather than replacing clinicians, digital twins are more likely to augment clinical decision-making, freeing professionals from routine tasks while enabling focus on complex, high-touch care requiring empathy, nuanced judgment, and contextual understanding that algorithms cannot replicate.[74] New roles such as clinical data scientists, digital twin specialists, or health informaticians may emerge to bridge technical and clinical domains, requiring educational programmes that blend clinical knowledge with computational skills.[105]

Healthcare workforce shortages, particularly acute in primary care and rural areas, might be partially mitigated through digital twins that extend specialist expertise through algorithmic decision support. A remote rural clinic with limited diagnostic capabilities might leverage digital twins to triage patients, identify those requiring specialist referral, and monitor chronic diseases between telemedicine consultations, effectively extending specialist reach while maintaining local care relationships.[110]

Health equity implications of digital twins remain uncertain and could unfold in contrasting directions. Optimistically, digital twins might democratise access to sophisticated care by encoding expert knowledge into algorithms deployable at scale, potentially reducing urban-rural and socioeconomic disparities in care quality.[111] Pessimistically, if digital twin access requires expensive infrastructure, specialist expertise, or out-of-pocket payments, the technology might exacerbate existing divides, creating a "two-tier" system where affluent patients receive algorithmic precision medicine while disadvantaged populations receive traditional care. Preventing this outcome requires deliberate policy choices prioritising equity, such as public investments in digital health infrastructure, reimbursement parity for digital and traditional services, and targeted deployment in underserved settings.

Patient autonomy and agency could be enhanced through digital twins that provide transparent information supporting informed decision-making, or potentially undermined if algorithmic predictions become overly deterministic or coercive. Maintaining appropriate human oversight, ensuring patients retain the right to decline algorithmic recommendations, and designing systems that support rather than replace clinical judgment will be essential to preserving patient-centred care principles.[112]

9. Limitations of the Review

This narrative review has several limitations that warrant acknowledgement. First, as a narrative rather than systematic review, our search strategy was not exhaustive, and article selection was not conducted according to predefined PRISMA protocols. While we attempted to capture the breadth of digital twin applications in healthcare through comprehensive literature exploration, we cannot claim to have identified every relevant publication, and selection bias may have influenced which studies were included. A systematic review with meta-analysis would provide more rigorous evidence synthesis, particularly for specific clinical applications where sufficient trials exist.

Second, the rapidly evolving nature of digital twin technology means that some information presented here may become outdated quickly. Many applications discussed remain in research or pilot stages, with limited real-world implementation and clinical validation. The review necessarily reflects the state of knowledge at the time of writing, but the field is advancing rapidly, and conclusions drawn here may require

revision as new evidence emerges.

Third, the review's breadth, while providing comprehensive overview, limits depth of analysis for any single application domain. More focused reviews examining specific clinical areas, such as digital twins in oncology or surgical planning, would provide greater technical detail and nuanced understanding than possible in a broad survey.

Fourth, we relied primarily on published literature, which may suffer from publication bias towards positive findings. Negative results, failed implementations, and lessons from unsuccessful digital twin projects are often under-reported but would provide valuable insights into barriers and pitfalls. Grey literature, conference proceedings, and industry reports were not systematically searched, potentially missing important developments.

Fifth, many digital twin implementations are proprietary commercial products with limited public documentation of technical details, validation methodologies, or performance metrics. This opacity makes comprehensive evaluation challenging and may result in an incomplete picture of the technology landscape.

Sixth, the review focuses primarily on high-income country contexts where most digital twin development has occurred, with limited discussion of applicability and challenges in low- and middle-income settings where healthcare infrastructure, data availability, and technical capacity differ substantially.

Finally, while we attempted to address ethical, legal, and social implications, these topics deserve more extensive analysis than provided here. Dedicated examinations of the ethical dimensions, regulatory challenges, or health equity implications of digital twins would complement the primarily technical focus of this review.

Despite these limitations, we believe this review provides a valuable synthesis of current digital twin applications in healthcare and a framework for understanding the technology's potential, challenges, and future directions.

10. Conclusion

Digital twin technology represents a paradigm shift in healthcare delivery, moving from population-based averages towards personalised, predictive, and preventive medicine grounded in individual patient physiology, behaviour, and circumstances. By creating dynamic virtual representations of patients, organs, or healthcare systems that continuously update through real-time data integration, digital twins enable simulation, optimisation, and decision support previously impossible with static models or clinical intuition alone.

Applications spanning personalised medicine, surgical planning, chronic disease management, hospital operations, and rehabilitation demonstrate the technology's versatility and transformative potential. Early evidence suggests that digital twins can improve clinical outcomes, reduce complications, optimise resource utilisation, and enhance patient engagement across diverse healthcare domains. The promise of testing interventions virtually before administering them to patients, predicting exacerbations before symptoms emerge, and rehearsing surgeries on patient-specific anatomical models holds immense appeal for clinicians seeking to improve care while reducing risks.

However, significant challenges remain before digital twins can achieve widespread clinical adoption and fulfil their transformative promise. Technical barriers including data integration complexity, computational demands, and model validation difficulties require continued research and development. Ethical concerns surrounding privacy, algorithmic bias, and consent necessitate thoughtful governance frameworks balancing innovation with patient protection. Regulatory uncertainty must be resolved through adaptive frameworks that ensure safety without stifling progress. Clinical workflow integration and trust-building demand human-centred design and iterative refinement based on frontline clinician feedback.

The path forward requires sustained collaboration across disciplinary boundaries, bringing together clinicians, data scientists, engineers, ethicists, regulators, patient advocates, and policymakers in genuine partnership. No single stakeholder group possesses the expertise to navigate the complex technical, ethical, and practical challenges ahead. Academic-industry partnerships can accelerate translation from research prototypes to clinical products, while public sector involvement ensures that equity, safety, and public interest remain central priorities.

Investment in infrastructure, including interoperability standards, high-performance computing resources, and digital health data networks, provides the foundation upon which digital twin applications will be built. Equally important are investments in education and workforce development, ensuring that current and future healthcare professionals possess the digital literacy to work effectively with algorithmic decision support while maintaining the clinical judgment, empathy, and ethical grounding that define excellent care.

Looking ahead, the integration of digital twin technology with other innovations, including artificial intelligence, genomics, advanced sensors, and telemedicine, will likely create synergies that amplify impact beyond what any single technology could achieve. Multi-scale digital twins spanning molecular to population levels, explainable AI providing transparent predictions, and population health applications informing policy decisions represent exciting frontiers for future exploration.

Ultimately, the success of digital twin technology will be measured not by technical sophistication but by its impact on patient outcomes, health equity, and healthcare sustainability. Technology is merely a tool; its value depends on whether it helps clinicians provide better care, enables patients to live healthier lives, and creates more equitable and efficient health systems. With thoughtful development, rigorous evaluation, and responsible implementation guided by patient-centred values, digital twin technology has the potential to substantially advance those goals, ushering in a new era of precision, predictive, and personalised healthcare.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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